DATA SCIENCE & ARTIFICIAL INTELLIGENCE



Calculus and Optimization

Integration

Lecture No. 03



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Recap of previous lecture



Topic

INTEGRATION

Topics to be Covered



Topic

INTEGRATION (Continued)

Standard Results :-

(1) Power formula : $\int x^a dx = \frac{x^{a+1}}{a+1}, a \neq -1$

(2) $\int \frac{1}{x} dx = \log x + C$

(3) $\int a^x dx = \frac{a^x}{\log_e a} + C$

(4) $\int e^x dx = e^x + C$

(5) $\int \sin x dx = -\cos x + C$

(6) $\int \cos x dx = \sin x + C$

(7) $\int \tan x dx = \log \sec x + C$

(8) $\int \cot x dx = \log \sin x + C$

(9) $\int \sec x dx = \log(\sec x + \tan x) + C$

(10) $\int (\operatorname{cosec} x) dx = \log(\operatorname{cosec} x - \cot x) + C$

(11) $\int \sec^2 x dx = \tan x + C$

(12) $\int \operatorname{cosec}^2 x dx = -\cot x + C$

(13) $\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2}(a \sin bx - b \cos bx)$

(14) $\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2+b^2}(a \cos bx + b \sin bx)$

(15) $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+C$

(16) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left(\frac{x-a}{x+a}\right)+C$

(17) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left(\frac{a+x}{a-x}\right)+C$

(18) $\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left(x+\sqrt{x^2+a^2}\right)+C$

(19) $\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left(x+\sqrt{x^2-a^2}\right)+C$

(20) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)+C$

(21) $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left(x+\sqrt{x^2+a^2}\right)+C$

(22) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left(x+\sqrt{x^2-a^2}\right)+C$

(23) $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+C$

(24) $\int \sec x \tan x\,dx = \sec x + C$, (25) $\int (\operatorname{cosec} x \cot x)\,dx = -\operatorname{cosec} x + C$

(23) Integration by parts $\Rightarrow$ $\int \underset{\text{I}}{u} \cdot \underset{\text{II}}{v}\,dx = u\int v\,dx - \int \left\{\frac{d}{dx}(u)\cdot\int v\,dx\right\}dx + c$



Inverse Trig func^n — Algebraic func^n — Exp. func^n

the letter which is coming 1st in the word ILATE should be assumed as 1st func^n 'u'

Shortcut - e $\int uv\,dx = uv_1 - u'v_2 + u''v_3 - u'''v_4 + \cdots\cdots$

where dash denotes diff & suffix denotes integration.

Q. $\int x \sin x\,dx = ?$ (u = x, v = $\sin x$)

$= x\int \sin x\,dx - \int \left\{\frac{d}{dx}(x)\cdot\int \sin x\,dx\right\}dx + C$

$= x(-\cos x) + \int \cos x\,dx + C$

$= -x\cos x + \sin x + C$

Q. $\int x^3 \sin x\,dx = ?$ = Time Consuming = Do yourself.

(ILATE-II) $\int x^3 \sin x\,dx = x^3(-\cos x) - 3x^2(-\sin x) + 6x(+\cos x) - 6(\sin x) + 0$

$= -x^3\cos x + 3x^2\sin x + 6x\cos x - 6\sin x + C$

(ILATE-II) $\int x \sin x\,dx = x(-\cos x) - 1(-\sin x) + 0$

$= -x\cos x + \sin x + C$

Q. $\int x^2 e^x dx = x^2(e^x) - 2x(e^x) + 2(e^x) + 0$

Take Care = This formula is useful only when 1st func^n is Polynomial.

Properties of Definite Integral →

(1) $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

(2) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

(3) $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

where $a < c < b$

(4) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

(5) $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Property (6) →

$$\int_0^{2a} f(x)\,dx = \boxed{\int_0^a \{f(x)+f(2a-x)\}\,dx} = \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(2a-x)=f(x) \\ 0, & \text{if } f(2a-x)=-f(x) \end{cases}$$

Property (7) →

$$\int_{-a}^{a} f(x)\,dx = \boxed{\int_0^a \{f(x)+f(-x)\}\,dx} = \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(-x)=f(x) \text{ i.e. } f(x) \text{ is Even} \\ 0, & \text{if } f(-x)=-f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$$

Q. HW $I = \int_0^{\pi/4} \log(1+\tan x)\,dx = ? = \int_0^{\pi/4} \log\left(1+\tan\left\{\frac{\pi}{4}-x\right\}\right)dx$

(a) $\pi/4$

(b) $\frac{\pi}{4}\log 2$

(c) 0

(d) $\frac{\pi}{8}\log 2$

$$= \int_0^{\pi/4} \log\left\{1+\left(\frac{1-\tan x}{1+1.\tan x}\right)\right\}dx \qquad \because \tan(A-B) = \frac{\tan A - \tan B}{1+\tan A \tan B}$$

$$= \int_0^{\pi/4} \log\left\{\frac{2}{1+\tan x}\right\}dx = \int_0^{\pi/4} \left[\log 2 - \log(1+\tan x)\right]dx$$

$$I = \int_0^{\pi/4} \log 2\,dx - \int_0^{\pi/4} \log(1+\tan x)\,dx$$

$$I = \log 2\,(x)_0^{\pi/4} - I$$

$$2I = \log 2\left(\frac{\pi}{4}-0\right) \Rightarrow I = \frac{\pi}{8}\log 2$$

Q-5 $I=\int_0^2 \frac{(x-1)^2 \sin(x-1)}{(x-1)^2+\cos(x-1)}\,dx = ? = \int_{-1}^{1} \frac{t^2 \sin t}{t^2+\cos t}\cdot dt = \int_{-1}^{1} f(t)\,dt$ (use Prop 7)

Put $(x-1)=t$ $\begin{cases} \text{At } x=0,\ t=-1 \\ \text{At } x=2,\ t=1 \end{cases}$

$dx = dt$

$f(t)=\frac{t^2 \sin t}{t^2+\cos t}$

$f(-t)=\frac{(-t)^2 \sin(-t)}{(-t)^2+\cos(-t)}$

$= \frac{t^2(-\sin t)}{t^2+\cos t} = -f(t)$ ⇒ $f(t)$ is an odd func^n

$= \int_0^1 [f(t)+f(-t)]\,dt$

$= \int_0^1 [f(t)-f(t)]\,dt$

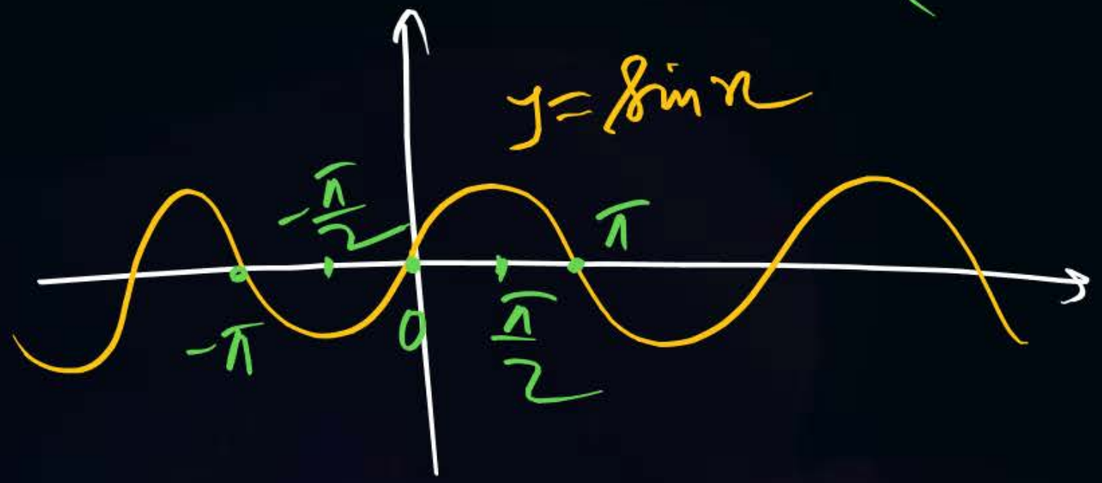
$= 0$

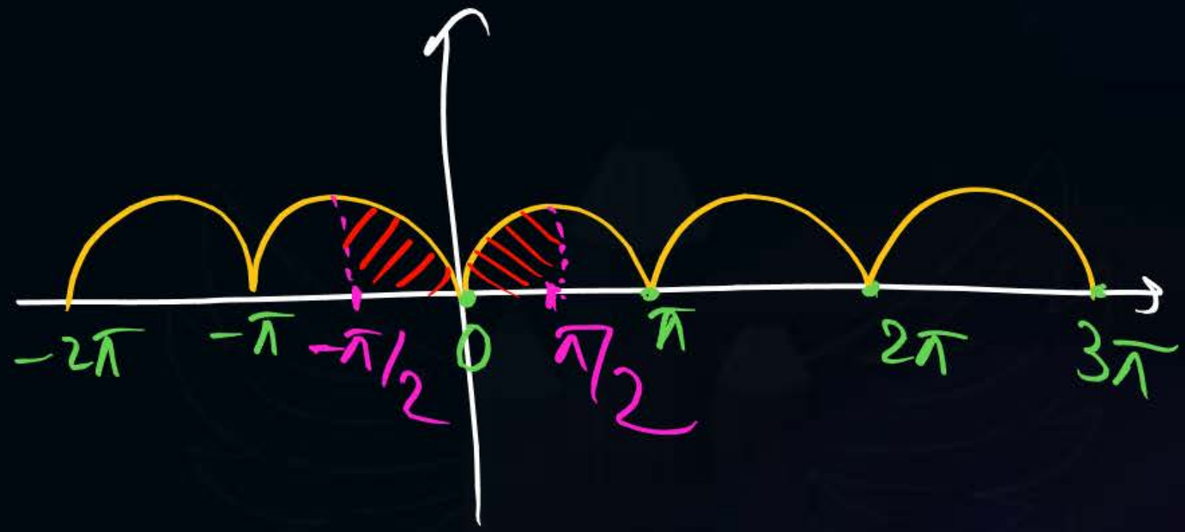
Q: $I=\int_{-\pi/2}^{\pi/2} |\sin x|\,dx = ?$ even func^{n}

Thm 7

$$=2\int_{0}^{\pi/2} |\sin x|\,dx = 2\int_{0}^{\pi/2} (+\sin x)\,dx$$

$$=2(-\cos x)_{0}^{\pi/2} = -2[0-1] = 2$$





Q. $I = \int_0^{2\pi} \cos^5 x\,dx = ? = 2\int_0^{\pi} \cos^5 x\,dx = 0$

Prop 6

Prop 6 Prop 6

$f(x) = \cos^5 x$

$f(-x) = \cos^5(-x)$

$= (\cos x)^5$

$= \cos^5 x$

$= f(x)$

i.e. $f(x)$ is Even func^n

Waste of time

$f(2\pi - x) = [\cos(2\pi - x)]^5 = (+\cos x)^5 = f(x)$

$f(\pi - x) = [\cos(\pi - x)]^5 = (-\cos x)^5 = -f(x)$

Q. $I = \int_{-3}^{3} \underbrace{\log\left(\frac{5-x}{5+x}\right)}_{\text{odd fun}^n} dx = ? = 0$ By Prop (7)

$f(x) = \log\left(\frac{5-x}{5+x}\right)$

$f(-x) = \log\left(\frac{5+x}{5-x}\right)$

$= \log\left(\frac{5-x}{5+x}\right)^{-1}$

$= -\log\left(\frac{5-x}{5+x}\right)$

$= -f(x)$

Q. $\int_0^{\pi/2} \log \sin x \, dx = ?$

(HW)

Ans: $-\frac{\pi}{2} \log 2$

Q. $\int_0^{\pi/2} \log \cos x \, dx = ?$

Ans: $-\frac{\pi}{2} \log 2$

(*) $\boxed{\int_0^{\pi/2} \log \sin x\, dx = \int_0^{\pi/2} \log \cos x\, dx = -\frac{\pi}{2}\log 2}$ Learn

Q. $I = \int_0^{\pi/2} \log(\tan x + \cot x)\, dx = \int_0^{\pi/2} \log\left\{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}\right\} dx = \int_0^{\pi/2} \log\left\{\frac{1}{\sin x \cos x}\right\} dx$

(a) $\frac{\pi}{2}\log 2$

(b) $\pi \log 2$

(c) $-\pi \log 2$

(d) $-\frac{\pi}{2}\log 2$

$= \int_0^{\pi/2} [\log 1 - \log(\sin x \cos x)]\, dx = -\int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right) dx$

$= -\int_0^{\pi/2} (\log \sin 2x - \log 2)\, dx = -I_1 + \int_0^{\pi/2} \log 2\, dx$

$I = -I_1 + \log 2\left(\frac{\pi}{2} - 0\right) = -I_1 + \frac{\pi}{2}\log 2$ ——— (1)

$$I_1 = \int_0^{\pi/2} \log \sin 2x \, dx$$

Put $2x = t$ ⟵ $x = 0, t = 0$; $x = \frac{\pi}{2}, t = \pi$

$$dx = \frac{dt}{2}$$

$$I_1 = \int_0^{\pi} \log(\sin t) \cdot \frac{dt}{2}$$

$$= \frac{1}{2} \int_0^{\pi} \log \sin t \, dt$$

$$= \frac{1}{2}\left[2\int_0^{\pi/2} \log \sin t \, dt\right]$$

$$I_1 = -\frac{\pi}{2}\log 2$$

Add By (1) $I = \left(-\frac{\pi}{2}\log 2\right) + \frac{\pi}{2}\log 2 = +\pi \log 2$

(M-II) $I = \dots = \int_0^{\pi/2} -(\log \sin x \cos x)\, dx$

$$= -\left\{\int_0^{\pi/2} \log \sin x \, dx + \int_0^{\pi/2} \log \cos x \, dx\right\}$$

$$= -\left\{-\frac{\pi}{2}\log 2 - \frac{\pi}{2}\log 2\right\}$$

$$= \pi \log 2$$

Leibnitz Rule of Diff under the Sign of Integration →

$$\frac{d}{dx}\int_{\phi(x)}^{\psi(x)} f(t)\,dt = \frac{d}{dx}(\psi)\,f(\psi) - \frac{d}{dx}(\phi)\,f(\phi)$$

eg $$\frac{d}{dx}\int_{x^2}^{x^3}\left(\frac{1}{\log t}\right)dt = ? = \frac{d}{dx}(x^3)\frac{1}{\log(x^3)} - \frac{d}{dx}(x^2)\frac{1}{\log(x^2)}$$

$$= \frac{3x^2}{3\log x} - \frac{2x}{2\log x} = \frac{x^2-x}{\log x}$$ Ans

Q: $\lim_{x\to 0}\left[\frac{\int_0^{x^2}\sqrt{4+t^3}\,dt}{x^2}\right] = ? \approx \frac{0}{0}$ form.

$$= \lim_{x\to 0}\left[\frac{\frac{d}{dx}\int_0^{x^2}\sqrt{4+t^3}\,dt}{\frac{d}{dx}(x^2)}\right]$$

$$= \lim_{x\to 0}\left[\frac{\frac{d}{dx}(x^2)\sqrt{4+(x^2)^3} - \frac{d}{dx}(0)\sqrt{4+(0)^3}}{2x}\right]$$

$$= \lim_{x\to 0}\left(\frac{2x\cdot\sqrt{4+x^6}-0}{2x}\right) = \lim_{x\to 0}\sqrt{4+x^6} = \sqrt{4+0} = 2$$

Q: Consider the func^n $f(x)=\int_0^x e^{-\left(\frac{x^2}{2}\right)}dx$ & let it's T.S.Exp about $x=0$ is given as $f(x)=a_0+a_1x+a_2x^2+a_3x^3+\cdots\cdots$ then $a_2=?$

Sol: we k.that in T.S.Exp, $a_2=\frac{f''(0)}{2!}=?$

$$f'(x)=\frac{d}{dx}\int_0^x e^{-\left(\frac{x^2}{2}\right)}\cdot dx=\frac{d}{dx}(x)e^{-\frac{x^2}{2}}-\frac{d}{dx}(0)e^{-\frac{0^2}{2}}$$

$$\Rightarrow f'(x)=e^{-\frac{x^2}{2}} \text{ \& } f''(x)=e^{-\frac{x^2}{2}}(-x)=-xe^{-\frac{x^2}{2}}$$

$$\text{So } f''(0)=0 \text{ So } a_2=\frac{f''(0)}{2}=0$$

BETA & GAMMA func^n

Properties of Gamma func^n →

(1) $\Gamma(n+1) = \begin{cases} n!, & n \in W \\ n\Gamma n, & n \in R^+ \end{cases}$

eg $\Gamma 5 = ? = \Gamma(4+1) = 4! = 24$

eg $\Gamma 6 = ? = \Gamma(5+1) = 5! = 120$

eg $\Gamma 2 = ? = \Gamma(1+1) = 1! = 1$

eg $\Gamma 8 = ? = 7!$ eg $\Gamma 4 = 3!$

$\Gamma 1/2 = \sqrt{\pi}$, $\Gamma 0 = N.D$

$\Gamma 3/2 = \Gamma(\frac{1}{2}+1) = \frac{1}{2}\Gamma\frac{1}{2} = \frac{\sqrt{\pi}}{2}$

$\Gamma 5/2 = ? = \Gamma(\frac{3}{2}+1) = \frac{3}{2}\Gamma\frac{3}{2} = \frac{3}{2}\cdot\frac{1}{2}\Gamma\frac{1}{2} = \frac{3\sqrt{\pi}}{4}$

$\Gamma 7/2 = ? = \Gamma(\frac{5}{2}+1) = \frac{5}{2}\Gamma\frac{5}{2} = \frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\Gamma\frac{1}{2}$

$\Gamma\frac{11}{2} = ? = \frac{9}{2}\cdot\frac{7}{2}\cdot\frac{5}{2}\cdot\frac{3}{2}\cdot\frac{1}{2}\cdot\Gamma\frac{1}{2}$

$\Gamma\frac{17}{4} = ? = \frac{13}{4}\cdot\frac{9}{4}\cdot\frac{5}{4}\cdot\frac{1}{4}\cdot\Gamma\frac{1}{4}$

② $\boxed{\Gamma n \cdot \Gamma(1-n) = \frac{\pi}{\sin n\pi}}$, $n \notin I$ ——— (1)

eg Evaluate $\Gamma\frac{1}{2}, \Gamma\frac{-1}{2}, \Gamma\frac{-3}{2}, \Gamma\frac{1}{4}\Gamma\frac{3}{4} = ?$

① Put $n = \frac{1}{2}$ in (1)

$$\Gamma\frac{1}{2} \cdot \Gamma\left(1-\frac{1}{2}\right) = \frac{\pi}{\sin\frac{\pi}{2}}$$

$$\left(\Gamma\frac{1}{2}\right)^2 = \frac{\pi}{(1)}$$

$$\Gamma\frac{1}{2} = \sqrt{\pi} \quad \text{Ans}$$

② Put $n = \frac{3}{2}$ in (1)

$$\Gamma\frac{3}{2} \cdot \Gamma\left(1-\frac{3}{2}\right) = \frac{\pi}{\sin\left(\frac{3\pi}{2}\right)}$$

$$\frac{1}{2}\Gamma\frac{1}{2} \cdot \Gamma\frac{-1}{2} = \frac{\pi}{(-1)}$$

$$\frac{\sqrt{\pi}}{2} \cdot \Gamma\frac{-1}{2} = -\pi$$

$$\Gamma\frac{-1}{2} = \frac{-2\pi}{\sqrt{\pi}} = -2\sqrt{\pi}$$

③ Put $n=\frac{5}{2}$ in (1),

$$\Gamma\frac{5}{2}\cdot\Gamma\left(1-\frac{5}{2}\right)=\frac{\pi}{\sin\frac{5\pi}{2}}$$

$$\left(\frac{3}{2}\cdot\frac{1}{2}\cdot\sqrt{\pi}\right)\cdot\Gamma\frac{-3}{2}=\frac{\pi}{(1)}$$

$$\Gamma\frac{-3}{2}=\frac{4\sqrt{\pi}}{3}$$

④ Put $n=\frac{1}{4}$ in (1)

$$\Gamma\frac{1}{4}\cdot\Gamma\left(1-\frac{1}{4}\right)=\frac{\pi}{\sin\left(\frac{\pi}{4}\right)}$$

$$\Gamma\frac{1}{4}\cdot\Gamma\frac{3}{4}=\frac{\pi}{1/\sqrt{2}}=\pi\sqrt{2}$$

(3) $$\int_0^{\pi/2} \sin^m\theta \cdot \cos^n\theta \, d\theta = \frac{\lceil\frac{m+1}{2} \cdot \lceil\frac{n+1}{2}}{2\lceil\frac{m+n+2}{2}}$$

Q. $I = \int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = ?$ Ans $= \frac{\pi}{\sqrt{2}}$

Sol: $= \int_0^{\pi/2} \sin^{\frac{1}{2}}\theta \cos^{-\frac{1}{2}}\theta \, d\theta$

$m = \frac{1}{2}, n = -\frac{1}{2}$

$$= \frac{\lceil\frac{\frac{1}{2}+1}{2} \cdot \lceil\frac{-\frac{1}{2}+1}{2}}{2\lceil\frac{\frac{1}{2}-\frac{1}{2}+2}{2}} = \frac{\lceil\frac{3}{4} \lceil\frac{1}{4}}{2\lceil 1} = \frac{\pi\sqrt{2}}{2(0!)} = \frac{\pi\sqrt{2}}{2} = \frac{\pi}{\sqrt{2}}$$ Ans

Q. $I = \int_0^{\pi/2} \cos^4 x \, dx = ?$

$$= \int_0^{\pi/2} \sin^0 x \cos^4 x \, dx = \frac{\lfloor \frac{0+1}{2} \lfloor \frac{4+1}{2}}{2 \lfloor \frac{0+4+2}{2}} = \frac{\lfloor \frac{1}{2} \lfloor \frac{5}{2}}{2 \lfloor 3}$$

$(m = 0, n = 4)$

$$= \frac{\sqrt{\pi} \cdot \left(\frac{3}{2} \cdot \frac{1}{2} \cdot \sqrt{\pi}\right)}{2\,(2!)} = \frac{3\pi}{16}$$

HW 8 / P 48

$$I = \int_0^{\pi/6} \cos^4(3\theta) \cdot \sin^3(6\theta)\, d\theta = ?$$

HINT: Put $3\theta = x$

Ans: $\frac{1}{15}$

(4) Def^n of Gamma func^n →

(2nd Eulerian Integral)

$$\boxed{\int_0^\infty e^{-x}\cdot x^{n-1}\,dx = \lceil n}$$

eg: $\int_0^\infty x^4 e^{-x}\,dx = ? = \int_0^\infty e^{-x}\cdot x^4\,dx = \int_0^\infty e^{-x}\cdot x^{5-1}\,dx = \lceil 5 = 4! = 24$

Qe $I = \int_0^\infty e^{-\sqrt{x}}\cdot x^{\frac{1}{4}}\,dx = ? = \int_0^\infty e^{-y}(y^2)^{\frac{1}{4}}\cdot 2y\,dy = 2\int_0^\infty e^{-y}\cdot y^{\frac{3}{2}}\,dy$

Put $\sqrt{x} = y \Rightarrow x = y^2$

$dx = 2y\,dy$

$= 2\int_0^\infty e^{-y}\cdot y^{\frac{5}{2}-1}\,dy = 2\cdot\lceil\frac{5}{2}$

$= 2\times\frac{3}{2}\times\frac{1}{2}\times\sqrt{\pi} = \frac{3\sqrt{\pi}}{2}$

Q. $I = \int_0^\infty e^{-x^2}\,dx = ?$

Put $x^2 = y \Rightarrow x = \sqrt{y}$

$dx = \frac{1}{2\sqrt{y}}\,dy$

$I = \int_0^\infty e^{-y} \cdot \frac{1}{2\sqrt{y}}\,dy$

$= \frac{1}{2}\int_0^\infty e^{-y} \cdot y^{-\frac{1}{2}}\,dy$

$= \frac{1}{2}\int_0^\infty e^{-y} \cdot y^{\frac{1}{2}-1}\,dy = \frac{1}{2}\Gamma\frac{1}{2} = \frac{\sqrt{\pi}}{2}$

HW Q. $I = \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{8}\right\} dx = ? = 1$

$= \int_0^\infty \frac{1}{\sqrt{2\pi}} e^{-\left(\frac{x^2}{8}\right)}\,dx = ?$

(5) Defⁿ of Beta funcⁿ (1st Eulerian Integral)

$$\int_0^1 x^{m-1}(1-x)^{n-1}\,dx = B(m,n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma m+n}$$

(1) Beta funcⁿ is symmetrical about m & n

i.e. $B(m,n) = B(n,m)$

(2) Relationship b/n Beta & Gamma funcⁿ is $B(m,n) = \frac{\Gamma m \cdot \Gamma n}{\Gamma m+n}$

(3) $$\int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}\,dx = B(m,n)$$

Q. $I=\int_0^1 x^5(1-x^3)^{10}dx=?$

$$=\int_0^1 (y^{1/3})^5(1-y)^{10}\cdot\frac{1}{3}y^{-\frac{2}{3}}dy$$

Put $x^3=y \Rightarrow x=y^{1/3}$

$dx=\frac{1}{3}y^{-2/3}\cdot dy$

$$=\frac{1}{3}\int_0^1 y^{\frac{5}{3}-\frac{2}{3}}\cdot(1-y)^{10}dy$$

$$=\frac{1}{3}\int_0^1 y^1\cdot(1-y)^{10}dy$$

$$=\frac{1}{3}\int_0^1 y^{2-1}\cdot(1-y)^{11-1}\cdot dy=\frac{1}{3}B(2,11)$$

$$=\frac{1}{3}\cdot\frac{\lceil 2\lceil 11}{\lceil 2+11}=\frac{1}{3}\cdot\frac{(1!)(10!)}{(12)!}=\frac{1}{396}$$

HW: $I=\int_0^2 x(8-x^3)^{1/3}\,dx=?$ Ans $\frac{16\pi}{9\sqrt{3}}$

Put $x^3=8y \Rightarrow x=2y^{1/3}$

$dx=2\cdot\frac{1}{3}y^{-\frac{2}{3}}\,dy$

At $x=0$, $y=0$; At $x=2$, $y=1$

$$I=\int_0^1 2y^{\frac{1}{3}}\cdot(8-8y)^{\frac{1}{3}}\cdot\frac{2}{3}y^{-\frac{2}{3}}\,dy$$

$$=\frac{8}{3}\int_0^1 y^{-\frac{1}{3}}(1-y)^{\frac{1}{3}}\,dy$$

$$I=\frac{8}{3}\int_0^1 y^{\frac{2}{3}-1}\cdot(1-y)^{\frac{4}{3}-1}\,dy$$

$$=\frac{8}{3}B\left(\frac{2}{3},\frac{4}{3}\right)=\frac{8}{3}\cdot\frac{\lfloor\frac{2}{3}\lfloor\frac{4}{3}}{\lfloor\frac{2}{3}+\frac{4}{3}}$$

$$=\frac{8}{3}\cdot\frac{\lfloor\frac{2}{3}\cdot\left(\frac{1}{3}\lfloor\frac{1}{3}\right)}{\lfloor 2}=\frac{8}{9}\cdot\frac{\lfloor\frac{1}{3}\lfloor 1-\frac{1}{3}}{1!}$$

$$=\frac{8}{9}\cdot\frac{\pi}{\sin\left(\frac{\pi}{3}\right)}=\frac{8}{9}\left(\frac{\pi}{\sqrt{3}/2}\right)=\frac{16\pi}{9\sqrt{3}}$$

Q. $I = \int_0^\infty \frac{x^8(1-x^6)}{(1+x)^{24}} = ? = \int_0^\infty \frac{x^8}{(1+x)^{24}}dx - \int_0^\infty \frac{x^{14}}{(1+x)^{24}}dx$

$$= \int_0^\infty \frac{x^{9-1}}{(1+x)^{9+15}}dx - \int_0^\infty \frac{x^{15-1}}{(1+x)^{15+9}}dx$$

$$= B(9,15) - B(15,9)$$

$$= B(9,15) - B(9,15)$$

$$= 0$$

$\because \int_0^\infty \frac{x^{m-1}}{(1+x)^{m+n}}dx = B(m,n)$

HW8 $I = \int_0^\infty \frac{dx}{1+x^4} = ?$ Ans: $\frac{\pi\sqrt{2}}{4}$

THANK - YOU

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